

Assignment-7 Functions in Python

Note : Use error handling to handle error gracefully

1. Write a function to print your name and age.
2. Write a function that accepts two numbers and returns their sum.
3. Write a function that takes a number and prints whether it is positive, negative, or zero.
4. Write a function to calculate the square of a given number.
5. Write a function to print all even numbers between 1 and 50.
6. Write a function that takes a number and prints its multiplication table.
7. Write a function to check whether a number is divisible by 5 or not.
8. Write a function that takes two strings and concatenates them.
9. Write a function to count how many times a word appears in a given sentence.
10. Write a function to return the length of a string without using the built-in `len()` function.
11. Write a function to count the number of vowels in a given string.
12. Write a function to check whether a string starts with a vowel.
13. Write a function to remove all spaces from a given string.
14. Write a function to replace all vowels in a string with `*`.
15. Write a function to count capital and small letters in a string.
16. Write a function to reverse a string passed by the user.
17. Write a function to check if a string is a palindrome or not.
18. Write a function to count the number of words in a given sentence.

19. Write a function to find the most repeated character in a string.
20. Write a function to convert all lowercase letters to uppercase without using `.upper()`.
21. Write a function to find the greatest element in a list.
22. Write a function to find the smallest element in a list.
23. Write a function to create a list by taking user inputs.
24. Write a function to remove duplicates from a list.
25. Write a function to count even and odd numbers in a list.
26. Write a function to find the sum of all elements in a list.
27. Write a function to sort a list without using the built-in `sort()` function.
28. Write a function to find the smallest string from a list of strings.
29. Write a function to find the factorial of a given number.
30. Write a function to check if a number is prime or not.
31. Write a function to find all prime numbers between 1 and 100.
32. Write a function to find the sum of digits of a number.
33. Write a function to calculate the Fibonacci series up to n terms.
34. Write a function to check if a number is an Armstrong number.
35. Write a function to create a text file and write "Hello, Python" in it.
36. Write a function to read content from a text file and display it.
37. Write a function to count the number of lines, words, and characters in a text file.
38. Write a function to append user input to an existing file.
39. Write a function that generates a random number between 1 and 100 and lets the user guess it.

40. Write a function to simulate rolling two dice and display their total.
Write a function using recursion to calculate the sum of natural numbers up to n.
41. Write a function to check if two strings are anagrams of each other.
42. Write a function that takes a sentence and returns the words in alphabetical order.
43. Write a function that uses list comprehension to find all even squares between 1 and 50.
44. Write a function that takes a list of numbers and returns a new list containing only unique elements.
45. Write a function using lambda to find the cube of a number.
46. Write a function that shuffles the elements of a list randomly.
47. Write a function that generates a random password of 8 characters using letters and digits.
48. Write a function to count how many numbers in a list are greater than the average of the list.
49. Write a function to display a menu-based simple calculator using `while True` loop and functions for each operation.
50. Write a function to return greatest substring in any sentence pass by user.